ECE551: Digital System Design and Synthesis

Fall 2026

Simulation Mechanism

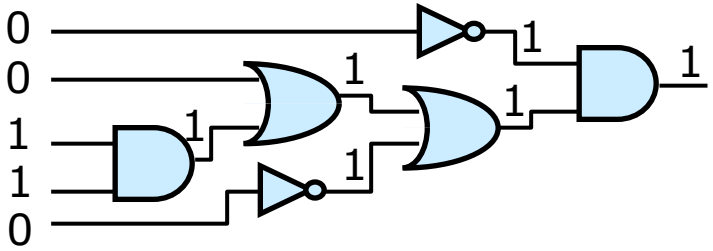
Timing Control for Simulation

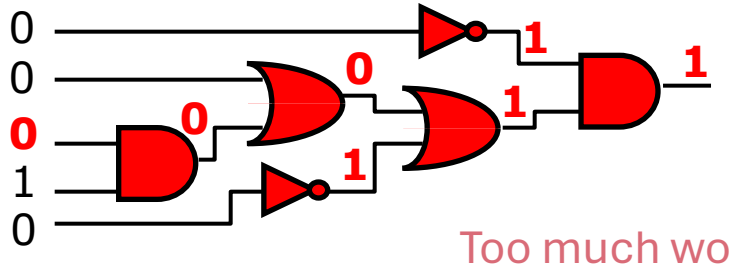
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Digital Simulation

- Software that models the behavior of your circuit
- Takes your design → model nets in the design
  - Keep track of their values at every point in time
- Takes your testbench → sequence of input values applied over time

- When inputs change, will need to recompute values





Too much work, too slow
Event-driven simulation instead

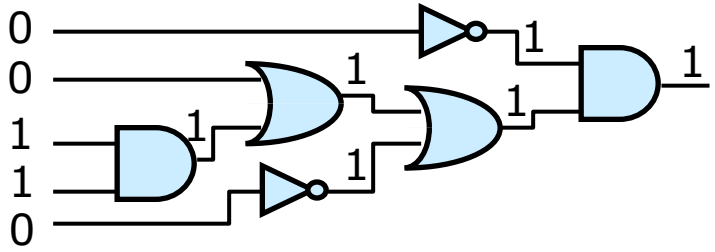
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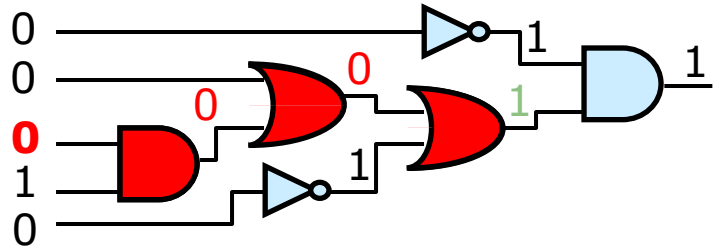
Event-Driven Simulation

- When an input to circuit changes, put it on a “changed” list
- Loop while the “changed” list isn’t empty
  - Remove a signal from the “changed” list
  - For each sink of the signal
    - Recompute its new output(s)
    - For any output(s) that have changed value, add that signal to the “changed” list
- When the “changed” list is empty:
  - Keep simulation results
  - Advance simulation time to the next stimulus (input) event

Digital Simulation

- When an input to circuit changes, put it on a “changed” list
- Loop while the “changed” list isn’t empty
  - Remove a signal from the “changed” list
  - For each sink of the signal
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Changed list: ~~0~~ ~~0~~ ~~0~~

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Functional vs Timing Simulation

- Functional simulation
  - Verifies logic correctness
  - Model with idealized delay
  - We spend most of the RTL design time here

- Timing simulation
  - Standard Delay Format (SDF) file contains propagation delays for gates and wires
  - Annotate netlist with delays and simulate to verify functional correctness
  - Vector dependent

Timing Controls for Simulation

- Can put “delays” in a Verilog design
  - Gates, wires, behavioral statements
- Delays are useful for simulation only!
  - Used to approximate “real” operation while simulating
  - Used to control testbench
  - Ignored by synthesis tool
- Types of delays
  - Inertial delays (Gates)
    - Gates need to have inputs held a certain time before output is accurate
    - Suppresses pulses shorter than delay amount
  - Transport delays (Nets)
    - “Time of flight” from source to sink
    - Short pulse transmitted

Delay Examples

- Inertial delay

```
and #4 (z_out, x_in, y_in);      // 4 unit inertial delay
assign #3 z_out = a & b;         // 3 unit inertial delay
```

- Transport delay

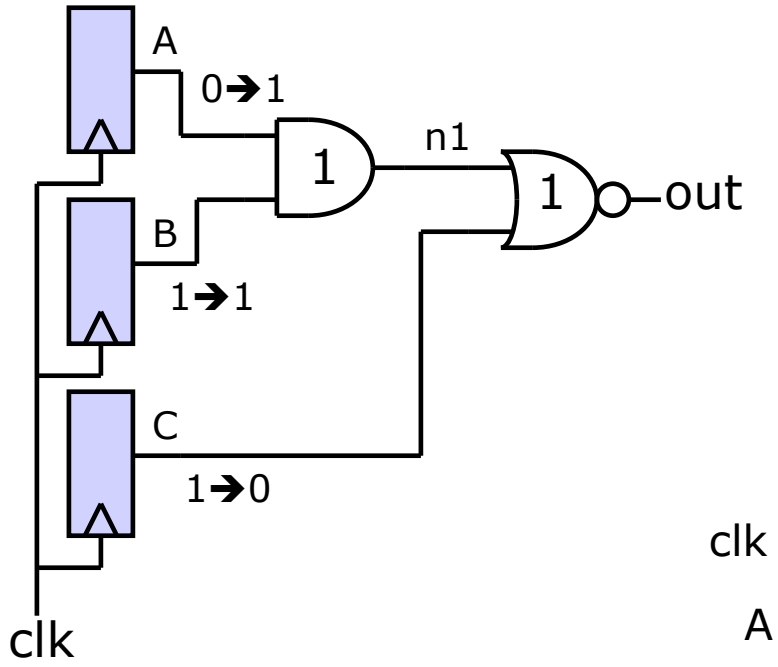
```
wire  #5  net_1;                 // 5 unit transport delay
```

- Used together

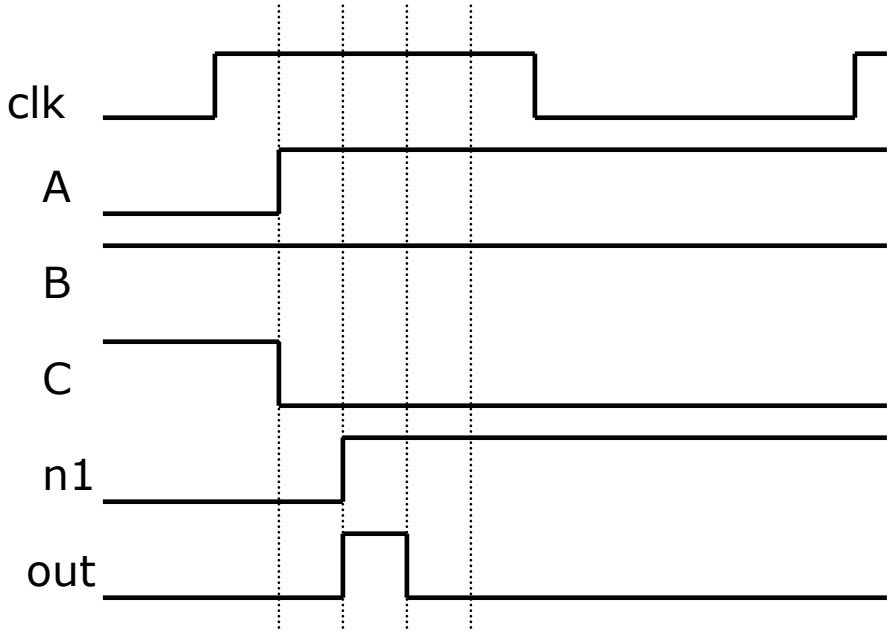
```
wire #2 z_out;                   // 2 unit transport delay
and #3 (z_out, x_in, y_in);      // 3 for gate, 2 for wire

wire #3 c;                       // 3 unit transport delay
assign #5 c = a & b;             // 5 for assign, 3 for wire
```

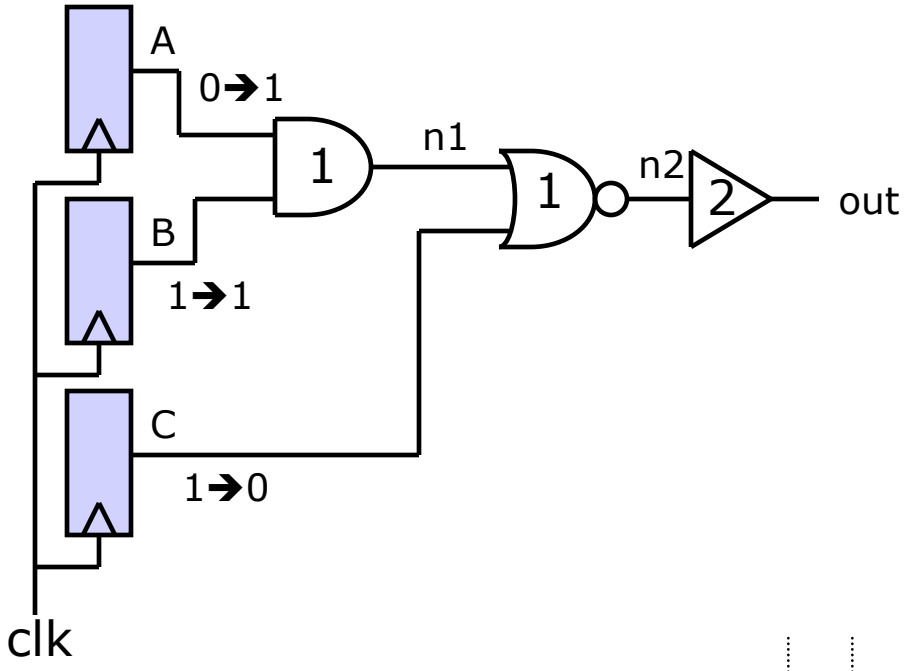
Combinational Logic Will Generate Pulses



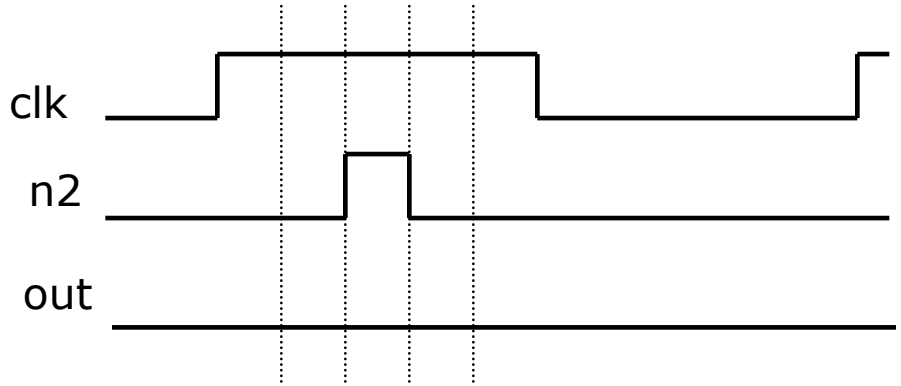
- Have 3 Flops with a unit delay clk2q
- Feeds simple combinational logic with unit gate delays
- A,B,C make the transitions shown
- What does “out” do?



Inertial Delay

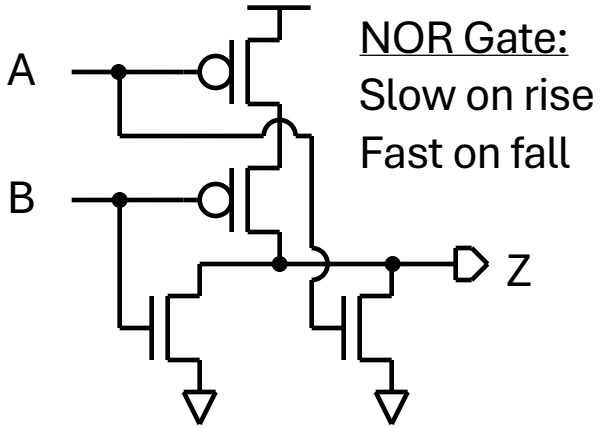


- If we add a buffer with a 2 unit delay to the end of this circuit, now what happens?
- Remember the inertial gate delay will filter any pulse shorter than the specified delay

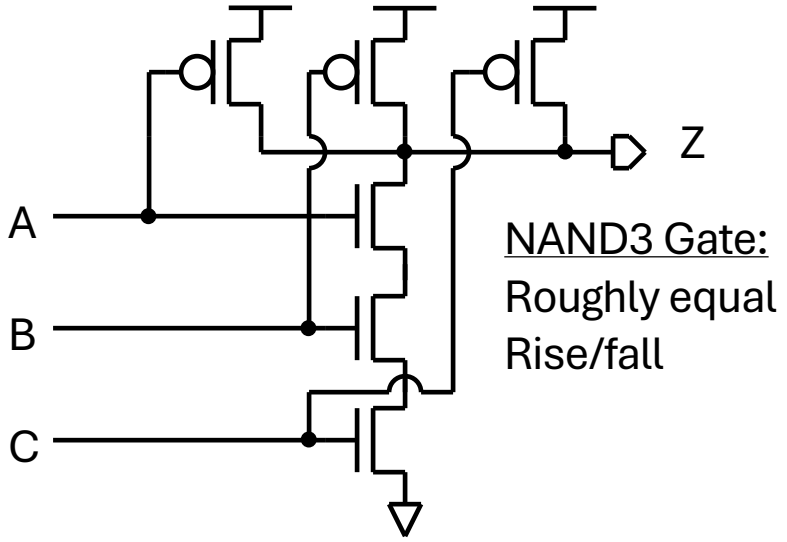


Gate Delay Variation

- Gates don’t have same delay in a rising transition vs falling
- Gate delays are not constant
  - Vary with PVT (Process, Voltage, Temperature)
  - You will hear reference to PVT corners
    - {fff, typ, sss} = {min, typ, max}



NOR Gate:
Slow on rise
Fast on fall



NAND3 Gate:
Roughly equal
Rise/fall

Gate Delay Variation

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    - {fff, typ, sss} = {min, typ, max}

- Analyze circuit speed using worst case operating condition
  - Is it fast enough?
    - Worst case = longest (max) delay
  - Race conditions?
    - Worst case = shortest (min) delay

Verilog Syntax for Delay Variations

- Rise/Fall/Turn off

```
not #(2,1) inv1(out,in);          // not gate with 2 unit rise and 1 unit fall

bufif1 #(1,1,2) tri1(out,in,en); // tri-state gate with unit rise/fall and 2 unit off
```

- Min/Typ/Max

```
and #(1:2:3) g1(out,a,b);         // 1 ns min, 2ns typical, 3ns max delay
```

- Can specify min/typ/max for rise and fall separately

```
and #(2:3:4, 1:2:3) g1(out,a,b); // gate has different rise,fall for min:typ:max
```

- Selection of timing set can is typically done in simulation environment

% verilog test.v +maxdelays

Invoke command line verilog engine (like Verilog XL) selecting the maxdelay time set.